19 March 2026

Verifiable Agent Conversation Records (Work in Progress)

AI Agent Security Side Meeting @ IETF 125 Shenzhen

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Overview

- Initial -00 @ datatracker
  https://datatracker.ietf.org/doc/draft-birkholz-verifiable-agent-conversations/
- Addressing technical interoperational requirements:
  - **Unifying/homogenizing** AI Agent **session logs** for post-processing
  - While **retaining** the original session logs as a **single source of truth**
- Main Purposes
  - "**AI Agent said it did X. Did it really?**"
  - **Detect & Prevent**: permission misuse, IPR or PII leaking, guardrail breaking, scheming, etc.
- Post-processing applications
  - **Auditability** and Accountability **after the fact**
  - **Quality Assurance** during d**evelopment and usage** of AI Agent models
  - Mid-term: (Near-)**Real-Time** AI Agent **Misbehavior Detection** (akin to SIEM systems)
  - **Regulation Compliance** (EU AI Act, ETSI TS 104 223, PCI DSS)

Requirement Frameworks & Intersections

Framework	Jurisdiction	Sector	Status
EU AI Act [EU_AI_ACT_2024]	EU	Cross-sector	In force Aug 2024
Cyber Resilience Act [EU_CRA_2024]	EU	Products with digital elements	In force Dec 2024
NIS2 Directive [EU_NIS2_2022]	EU	Essential/important entities	Transposed Oct 2024
ETSI TS 104 223 [ETSI_TS_104223_2025]	EU/International	AI systems	Published Apr 2025
SOC 2 Trust Services Criteria [AICPA_TSC_2017]	US/International	Service organizations	Active
FedRAMP [NIST_SP_800_53_R5] [OMB_M_21_31]	US	Federal cloud services	Active
PCI DSS [PCI_DSS_V4_0_1]	International	Payment card industry	Mandatory Mar 2025
ISO/IEC 42001 [ISO_IEC_42001_2023]	International	AI management systems	Published 2023
FFIEC IT Handbook [FFIEC_IT_HANDBOOK_2024]	US	Financial institutions	Updated 2024
BSI AI Finance Test Criteria [BSI_AI_FINANCE_2024]	Germany	Financial sector AI	Published 2024
NIST AI 100-2 [NIST_AI_100_2_E2025]	US	Cross-sector	Published 2025

1. Automatic Event Logging
2. Timestamp Requirements
3. Actor Identification
4. Action/Event Type Recording
5. Input/Output Recording (AI-Specific)
6. Retention Period Requirements
7. Tamper-Evidence and Integrity Protection
8. Incident Response Support
9. Anomaly and Risk Detection Support
10. Human Oversight Enablement
11. Traceability and Reproducibility

Content by Example (I)

```
verifiable-agent-record = {
    version: tstr
    id: tstr
    session: session-trace
    ? created: abstract-timestamp
    ? file-attribution: file-attribution-record
    ? vcs: vcs-context
    ? recording-agent: recording-agent
    * tstr => any
}
```

- **session-trace**: the full conversation trace including all entries, tool calls, reasoning steps, and system events.
- **file-attribution**: structured data about which files were modified and which line ranges were written by the agent
- **vcs**: version control metadata at the record level (repository, branch, revision)
- **recording-agent**: identifies the tool or agent that generated this record, as distinct from the agent that conducted the conversation

Content by Example (I)

```
session-trace = {
    ? format: tstr
    session-id: session-id
    ? session-start: abstract-timestamp
    ? session-end: abstract-timestamp
    agent-meta: agent-meta
    ? environment: environment
    entries: [* entry]
    * tstr => any
}
```

- **format**: Classifies the session style, such as "interactive" (human-in-the-loop) or "autonomous" (fully automated). Informative only; does not change the structure
- **agent-meta**: metadata about the agent and model that conducted this conversation
- **environment**: Execution environment context, such as working directory and version control state
- **entries**: an ordered array of conversation entries.

Next Steps

- Increase applicability and interoperability by **testing against multiple magnitudes of session logs** from all major frontier models
- **Unify, concretize, and simply** some of the existing and resulting semantics
- Adding **COSE Hash Envelope** support to add concise references to the original records as a single source of truth (https://datatracker.ietf.org/doc/draft-ietf-cose-hash-envelope/)
- Cooperation with frontier labs for **native VACR production** (rendering it the single source of truth)
- Adding complementary I-Ds to increase:
  - **believability/authenticity** of original session logs and records (RATS)
  - **transparency/non-repudiation** of VACRs for auditability (SCITT)

- Questions? Thoughts?